



The Revised Neo Personality Inventory and the MMPI-2 Psychopathology Five in the prediction of aggression

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Abstract

As suggested by Ben-Porath and Waller (Ben-Porath, Y. S., & Waller, N. G. (1992). “Normal” personality inventories in clinical assessment: general requirements and the potential for using the NEO personality inventory. *Psychological Assessment*, 4, 14–19], the incremental validity of the Revised NEO Personality Inventory [NEO-PI-R; Costa, P. T., & McCrae, R. R. (1992b). *The revised NEO personality inventory (NEO-PI-R) and NEO five factor inventory (NEO-FFI) professional manual*. Odessa, FL: Psychological Assessment Resources] domain scales over the Minnesota Multiphasic Personality Inventory-2 Personality Psychopathology Five [MMPI-2 PSY-5; Harkness, A. R., McNulty, J. L., & Ben-Porath, Y. S. (1995). The personality five (PSY-5): constructs and MMPI-2 scales. *Psychological Assessment*, 7, 164–114] scales for predicting aggression was assessed in the present study using a sample of 234 introductory psychology students (155 females, 79 males). Results indicated that the NEO-PI-R domain scales were able to add significantly to the prediction of several facets of aggression over and above the PSY-5 scales, providing evidence for the incremental validity of the instrument. However, the practical importance of the findings remain open to debate. Implications of the present findings for the use of normal-range personality instruments in clinical settings are discussed. © 2001 Elsevier Science Ltd. All rights reserved.

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1. Introduction

In recent years there has been great interest in the relationship between normal personality functioning and psychopathology (e.g. Maher & Maher, 1994), especially in the context of the

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five-factor model of personality (Ben-Porath & Waller, 1992; McCrae, 1991; Widiger & Trull, 1992). A great debate has emerged from this area concerning whether or not instruments assessing normal personality traits are useful for clinical assessment (Ben-Porath & Waller; Butcher & Rouse, 1996; Costa & McCrae, 1992a). Costa & McCrae have vehemently argued for the usefulness of measures of the “Big Five” (i.e. Neuroticism, Extraversion, Openness, Agreeableness, Conscientiousness) in clinical contexts, especially their own NEO Personality Inventory (NEO-PI; Costa & McCrae, 1985). More specifically, McCrae offered three reasons why such measures would be of interest to clinicians: (1) they measure emotional, interpersonal, and motivational styles that are relevant to the diagnosis of a wide range of other disorders; (2) they offer a comprehensive picture of the individual that cannot be obtained with most clinically-oriented instruments; and (3) they provide supplementary information that may be useful in selecting treatment and anticipating the course of therapy.

Other psychologists have not been convinced that measures of normal personality traits such as the NEO-PI and its successor the Revised NEO Personality Inventory (NEO-PI-R; Costa & McCrae, 1992b) can add any relevant information to current methods of clinical assessment (Ben-Porath & Waller, 1992; Butcher & Rouse, 1996). Along these lines, Ben-Porath and Waller (1992) had this to say about future investigations into the potential usefulness of the NEO-PI in clinical contexts:

Perhaps the most important of these is the need to establish whether the NEO-PI contributes incrementally to the procurement of diagnostic information beyond that which is obtained from current clinical measures. In this context, we would add that establishment of incremental validity cannot be accomplished by the sole presentation of zero-order correlations between the NEO-PI and clinical measures. Rather, the use of multivariate techniques, such as hierarchical regression analyses with clinically relevant extratest data serving as dependent variables, is called for (p. 17).

Ben-Porath and Waller went on to suggest that the use of measures like the NEO-PI in clinical assessments should not only make an incremental contribution that is clinically significant, but must also be judged as worthy of the extra time to administer and interpret the test.

One of the obvious benefits of using normal personality instruments such as the NEO-PI is that they are often based on dimensional models as opposed to the traditional categorical models commonly associated with clinical assessment. Recently, Harkness and McNulty (1994) have presented a new dimensional system for describing personality and its disorders, which they referred to as the Personality Psychopathology Five (PSY-5). The PSY-5 involves five basic constructs labeled Aggressiveness, Psychoticism, Constraint, Negative Emotionality/Neuroticism, and Positive Emotionality/Extraversion that are believed to make up the structure of normal and disordered personality. Harkness, McNulty and Ben-Porath (1995) developed scales designed to assess the PSY-5 constructs based on the Minnesota Multiphasic Personality Inventory-2 (MMPI-2; Butcher, Dahlstrom, Graham, Tellegen & Kaemmer, 1989), referred to as the MMPI-2 PSY-5. In reference to this instrument and the dimensional model, Butcher and Rouse (1996) stated that “those who are dissatisfied with categorical models in assessment of abnormal behavior and are seeking dimensional replacements . . . may find the PSY-5 more appropriate than the less pathologically oriented items in the Big Five” (p. 100).

Following Ben-Porath and Waller's (1992) suggestion concerning the suitability of normal personality measures in clinical contexts, the purpose of the present study was to examine the incremental validity of the NEO-PI-R over the MMPI-2 PSY-5 for predicting dimensions of aggression. Aggression was chosen as a criterion in this study because of its obvious clinical relevance (e.g. Kernberg, 1992). In addition, some researchers have suggested that basic personality traits (e.g. neuroticism, agreeableness) may act as antecedents to aggressive behavior (Caprara, Barbaranelli, Pastorelli & Perugini, 1994; Geen & Donnerstein, 1998). The Aggression Questionnaire (Buss & Perry, 1992), which has subscales for assessing four dimensions of aggression (Physical Aggression, Verbal Aggression, Anger, and Hostility), was selected to measure aggression. The MMPI-2 PSY-5 was selected as a clinical measure in the present investigation for two reasons: (1) it is based on the MMPI-2 which is a typical measure used in clinical contexts; and (2) it is a likely substitute for measures of normal personality because it is based on a dimensional model. If the NEO-PI-R domain scales are able to predict additional variance in the aggression scales over and above the MMPI-2 PSY-5 scales, this would provide some evidence for the potential usefulness of the NEO-PI-R as an additional measure to be added to clinical test batteries. Past research has indicated that the NEO-PI can add significantly to the prediction of personality disorders over and above broad measures of depression and anxiety (Trull, Useda, Costa & McCrae, 1995). Thus, the present study will add to the existing literature on this topic.

Although human aggression has been widely studied as a behavioral phenomenon (Buss, 1961; Feshbach & Zagrodzka, 1997; Geen, 1990; Geen & Donnerstein, 1983, 1998; Kernberg, 1992; Silverberg & Gray, 1992), much less theory and research has been devoted to understanding the role personality variables play as potential antecedents to aggressive behavior (Geen & Donnerstein, 1998). However, past research seems to suggest that aggression is broadly related to several basic personality/temperament traits. It is important to note first that some theorists view aggression as a dispositional trait in and of itself that gives rise to or predicts observed instances of aggressive behavior (e.g. Harkness et al., 1995; Tellegen, 1982), while others have theorized that a variety of personality traits operate as antecedents to aggressive behavior (e.g. Caprara, 1986; Caprara, Barbaranelli & Comrey, 1992; Caprara, Barbaranelli, & Zimbardo, 1996). Not surprisingly, a positive relationship between neuroticism/emotional instability and aggression has been observed in several studies (Buss & Perry, 1992; Caprara, Cinanni, D'Imperio, Passerini & Travaglia 1985; Harkness et al., 1995; Lynn, Hampson & Agahi, 1989). Other studies have observed a positive relationship between aggression and psychoticism (Lynn et al.; Rushton, Fulker, Neale, Nias & Eysenck, 1989). In addition to emotional instability, Caprara and associates have identified irritability and dissipation-rumination (i.e. decrease or increase in the likelihood of aggressing with the passage of time) as potential dispositional antecedents of aggressive behavior (Caprara, 1986; Caprara et al.). Finally, Buss and Perry observed positive relationships between aggression and impulsiveness, assertiveness, and competitiveness, as well as activity (for males only).

This brief review of past research concerning the relationship between basic personality traits and aggression provides some rationale for making specific hypotheses for the scales used in the present investigation. It was hypothesized that strong positive relationships would be observed between the aggression scales and PSY-5 Aggressiveness and Negative Emotionality and NEO-PI-R Neuroticism, and a strong negative relationship between aggression and NEO-PI-R Agreeableness. The predicted relationship between PSY-5 Aggressiveness and dimensions of aggression is quite

obvious and brings up methodological issues that will be discussed later. The positive relationship between aggression and neuroticism/negative emotionality has been demonstrated previously in the research literature (e.g. Lynn et al. 1989), and is expected to be particularly strong for the aggression facets of anger and hostility (as measured by the Aggression Questionnaire [AQ]). Theoretically, it seems appropriate to expect a negative association between agreeableness and aggression, and Buss and Perry's (1992) finding of a positive relationship between aggression and competitiveness offers some empirical support for this hypothesis.

In addition to these expected strong associations with aggression, other more modest associations were expected for some of the remaining personality scales. First, a positive relationship was expected between the aggression scales and PSY-5 Psychoticism. Although positive correlations between psychoticism and aggression have been obtained in the literature, Harkness et al. (1995) have claimed that their PSY-5 Psychoticism scale is not closely related to Eysenck's (Eysenck & Eysenck, 1985) popular conceptualization of psychoticism, but that it assesses degree of reality contact. However, they did obtain a positive correlation between PSY-5 Psychoticism and the Aggression scale of the Multidimensional Personality Questionnaire (MPQ; Tellegen, 1982), providing some support for the present hypothesis. In addition to psychoticism, PSY-5 Constraint was hypothesized to be negatively related to aggression. One important feature of trait constraint is impulsivity which has been shown in past research to have a positive relationship with aggression (Buss & Perry, 1992). Additionally, Harkness et al. observed a positive correlation between PSY-5 Constraint and MPQ Aggression, providing further support for this hypothesis. Finally, weaker negative relationships were expected between the aggression scales and NEO-PI-R Conscientiousness and Extraversion as well as PSY-5 Positive Emotionality, and no association was expected for NEO-PI-R Openness.

2. Method

2.1. Participants

Participants consisted of 234 undergraduate students (155 females, 79 males) from a midsized Midwestern university. The mean age of the participants was 19.8 (S.D. = 4.1) and 63.4% were Caucasian. Partial course credit was given for participation in the study.

2.2. Instruments

2.2.1. The AQ

The 29-item AQ (Buss & Perry, 1992) contains four factor analytically derived subscales: Physical Aggression (9 items), Verbal Aggression (5 items), Anger (7 items), and Hostility (8 items). The questionnaire was intended to assess multiple components of aggression and to improve upon the dated Buss-Durkee Hostility Inventory (Buss & Durkee, 1957). A 5-point Likert scale is used for the questionnaire ranging from "extremely uncharacteristic of me" to "extremely characteristic of me". Internal consistency reliabilities (Cronbach's alpha) for the aggression scales reported by Buss & Perry using a college sample of 1253 subjects were 0.85, 0.72, 0.83, 0.77, and 0.89 for Physical Aggression, Verbal Aggression, Anger, Hostility, and Total Aggression, respectively.

Test-retest reliabilities (9-week interval) for the scales using a college sample of 372 subjects were 0.80, 0.76, 0.72, 0.72, and 0.80 for the same five scales.

2.2.2. *The NEO-PI-R*

The NEO-PI-R (Costa & McCrae, 1992b) contains 240 items with five domain scales assessing broad personality traits of the five-factor model of personality: Neuroticism (N), Extraversion (E), Openness to Experience (O), Agreeableness (A), and Conscientiousness (C). The five domain scales can be further reduced into thirty facet scales (six for each domain scale) containing eight items each. A 5-point Likert scale is used for the NEO-PI-R ranging from “strongly disagree” to “strongly agree”. Internal consistency reliabilities (Cronbach’s alpha) reported by Costa and McCrae using a sample of 1539 respondents were 0.92, 0.89, 0.87, 0.86, and 0.90 for N, E, O, A, and C, respectively. Test-retest reliabilities (3-month interval) computed using a sample of 208 college students were 0.79, 0.79, 0.80, 0.75, and 0.83 for the five domain scales.

2.2.3. *The MMPI-2 PSY-5*

Harkness et al. (1995) developed the 139 item MMPI-2 PSY-5 to assess the psychopathological aspects of personality through five broad constructs: Aggressiveness (18 items), Psychoticism (25 items), Constraint (29 items), Negative Emotionality/Neuroticism (33 items), and Positive Emotionality/Extraversion (34 items). The authors made an effort to include items tapping both psychopathological and normal personality functioning, thus, the instrument can be administered to normal or clinical samples. A rational method of item analysis was used to construct scales assessing the five constructs based on items from the MMPI-2 (Butcher et al. 1989). The MMPI-2 uses a true-false item response format. Internal consistency reliabilities (Cronbach’s alpha) computed using a college sample ($n = 2,928$) were 0.71, 0.74, 0.69, 0.84, and 0.74 for Aggressiveness, Psychoticism, Constraint, Negative Emotionality/Neuroticism, and Positive Emotionality/Extraversion, respectively. No reports of test-retest reliability were given.

2.3. *Procedure*

Participants were first given an informed consent form, a set of written instructions, and brief verbal instructions. Next, the AQ, the NEO-PI-R, and the MMPI-2 PSY-5 were administered to the participants in small groups over the course of approximately 10 months. The order of presentation of instruments was counterbalanced across sessions to minimize order effects. Finally, participants were debriefed and thanked for their participation.

2.4. *Data analysis*

After calculating means, standard deviations, and internal consistency reliabilities for each scale, and computing the intercorrelations among the personality and aggression scales, five hierarchical multiple regression analyses were conducted in order to test the incremental validity of the NEO-PI-R domain scales over and above the PSY-5 scales for predicting aggression. More specifically, the regression analyses were conducted using the five AQ scales as separate criteria, with the five PSY-5 scales entered in the first step of each analysis and the five NEO-PI-R scales entered in the second step. Similar analyses were conducted with the NEO-PI-R scales entered in

the first step and the PSY-5 scales entered in the second to determine if the PSY-5 scales could add additional variance to the prediction of aggression. Since the PSY-5 Aggressiveness scale was expected to be highly related to the AQ scales (i.e. construct overlap), an additional set of hierarchical analyses were conducted using the AQ scales as criteria and the same two sets of predictors, except that the PSY-5 Aggressiveness scale was dropped from the analyses.

3. Results

An alpha level of 0.05 was used for all statistical tests. The means and standard deviations for all scales are reported in Table 1 for both males and females. Table 1 also provides estimates of internal consistency reliability (Cronbach's alpha) for all scales used in the study. Cronbach's alphas for the MMPI-2 PSY-5 and NEO-PI-R domain scales were generally comparable with established reliability estimates for the scales, however, alphas obtained for the PSY-5 scales were slightly higher than those reported by Harkness et al. (1995) for a sample of college students. Cronbach's alphas for the AQ scales were also comparable to published estimates. Given that subsequent regression analyses include the personality scales in a joint fashion, correlations of the NEO-PI-R scales with the MMPI-2 PSY-5 scales are provided in Table 2.

Table 1

Means, standard deviations, and internal consistency reliabilities (Cronbach's Alpha) for males and females on violence-related measures and personality scales^a

Scale	α	Females ($n = 155$)		Males ($n = 79$)	
		Mean	S.D	Mean	S.D.
<i>AQ Scales</i>					
Physical Aggression	0.85	22.41	8.25	25.49	7.70
Verbal Aggression	0.71	15.32	4.41	15.46	3.85
Anger	0.79	17.50	5.09	16.35	5.04
Hostility	0.78	21.32	6.83	19.97	5.70
Total	0.91	76.55	20.83	77.28	18.66
<i>NEO-PI-R Scales</i>					
Neuroticism	0.89	95.45	22.84	86.38	19.38
Extraversion	0.87	123.21	20.66	111.54	16.56
Openness	0.88	117.11	19.64	114.49	21.74
Agreeableness	0.87	110.07	19.23	105.19	21.04
Conscientiousness	0.89	110.76	20.03	108.28	22.20
<i>MMPI PSY-5 Scales</i>					
Aggressiveness	0.74	9.31	3.58	10.11	3.47
Psychoticism	0.80	6.99	4.20	7.19	4.57
Constraint	0.80	15.42	3.79	12.46	4.82
Negative Emotionality	0.85	14.16	5.67	11.49	5.94
Positive Emotionality	0.88	24.22	5.40	23.14	4.50

^a AQ, Aggression Questionnaire; NEO-PI-R, Revised NEO Personality Inventory; MMPI-2 PSY-5, Minnesota Multiphasic Personality Inventory-2 Psychopathology Five.

Table 3 provides correlations between the AQ scales and the two sets of personality scales. Of the NEO-PI-R domain scales, the subscales of the AQ as a set showed the most substantial correlations with NEO-PI-R Neuroticism and Agreeableness, such that higher aggression was associated with higher neuroticism and lower agreeableness. For the PSY-5 scales, higher scores on the AQ subscales were broadly associated with higher scores on the Aggressiveness, Negative Emotionality, and Psychoticism scales, and a lower score on the Constraint scale. Finally, small to moderate correlations were observed between AQ scales and NEO-PI-R Extraversion and Conscientiousness, as well as PSY-5 Positive Emotionality. The magnitude and direction of these correlations were largely as hypothesized.

Tables 4 and 5 display the results of the hierarchical regression analyses conducted to test the incremental validity of the five NEO-PI-R domain scales over the five MMPI-2 PSY-5 scales for predicting the five AQ scales (i.e. the four facet scales and the total scale score). The adjusted multiple *Rs* in the first step of the analysis with the five PSY-5 scales were statistically significant

Table 2

Correlations of Revised NEO Personality Inventory (NEO-PI-R) Scales with Minnesota Multiphasic Personality Inventory-2 Psychopathology Five (MMPI-2 PSY-5) Scales

MMPI-2 PSY-5 Scales	NEO-PI-R Scales ^a				
	N	E	O	A	C
Aggressiveness	0.11	0.04	0.10	–0.53	0.09
Constraint	–0.11	–0.02	–0.25	0.29	0.30
Psychoticism	0.38	–0.15	0.04	–0.32	–0.11
Negative Emotionality	0.71	–0.18	–0.08	–0.27	–0.15
Positive Emotionality	–0.40	0.60	0.21	0.29	0.28

^a NEO-PI-R Scales: N, Neuroticism; E, Extraversion; O, Openness to Experience; A, Agreeableness; C, Conscientiousness. All values above 0.13 are significant at the 0.05 alpha level and values above 0.30 are in italics.

Table 3

Correlations of aggression scales with personality scales^a

AQ Scales	NEO-PI-R Scales ^b					MMPI-2 PSY-5 Scales ^c				
	N	E	O	A	C	Ag	Co	Ps	NE	PE
Physical Aggression	0.26	–0.16	–0.02	–0.47	–0.18	0.56	–0.39	0.33	0.30	–0.18
Verbal Aggression	0.20	–0.02	0.13	–0.48	–0.12	0.53	–0.28	0.26	0.25	0.03
Anger	0.51	–0.17	–0.01	–0.49	–0.24	0.55	–0.20	0.40	0.56	–0.21
Hostility	0.61	–0.25	–0.09	–0.32	–0.18	0.33	–0.11	0.40	0.58	–0.37
AQ Total	0.49	–0.20	–0.01	–0.54	–0.23	0.60	–0.31	0.43	0.52	–0.25

^a AQ, Aggression Questionnaire; NEO-PI-R, Revised NEO Personality Inventory; MMPI-2 PSY-5, Minnesota Multiphasic Personality Inventory-2 Psychopathology Five. All values above 0.13 are significant at the 0.05 alpha level and values above 0.30 are in italics.

^b NEO-PI-R Scales: N, Neuroticism; E, Extraversion; O, Openness to Experience; A, Agreeableness; C, Conscientiousness.

^c MMPI-2 PSY-5 Scales: Ag, Aggressiveness; Co, Constraint; Ps, Psychoticism; NE, Negative Emotionality; PE, Positive Emotionality.

Table 4

Incremental validity of Revised NEO Personality Inventory (NEO-PI-R) over Minnesota Multiphasic Personality Inventory-2 Psychopathy Five (MMPI-2 PSY-5; with and without Aggressiveness Scale) for predicting aggression^a

Aggression Scale	MMPI-2 PSY-5		MMPI-2 PSY-5 + NEO-PI-R		Incremental validity	
	Multiple <i>R</i>	<i>F</i> (5, 229)	Multiple <i>R</i>	<i>F</i> (10, 224)	<i>R</i> ² change	<i>F</i> (5, 224) change
<i>With PSY-5 Aggressiveness:</i>						
Physical Aggression	0.64	33.21*	0.67	18.72*	0.02	2.86*
Verbal Aggression	0.55	20.83*	0.61	14.78*	0.07	6.31*
Anger	0.69	41.97*	0.75	29.99*	0.09	9.80*
Hostility	0.69	43.24*	0.74	29.36*	0.07	8.37*
Total Score	0.76	65.54*	0.81	44.84*	0.08	10.92*
	Multiple <i>R</i>	<i>F</i> (4, 230)	Multiple <i>R</i>	<i>F</i> (9,225)	<i>R</i> ² change	<i>F</i> (5,225) change
<i>Without PSY-5 Aggressiveness:</i>						
Physical Aggression	0.51	21.55*	0.58	14.07*	0.07	6.14*
Verbal Aggression	0.38	11.10*	0.56	12.54*	0.16	11.63*
Anger	0.60	32.71*	0.69	23.91*	0.12	11.05*
Hostility	0.68	50.52*	0.73	29.70*	0.07	7.35*
Total score	0.66	44.86*	0.74	31.79*	0.11	12.33*

^a All Multiple *R*s are adjusted

Table 5

Beta weights for regression analyses examining the incremental validity of Revised NEO Personality Inventory (NEO-PI-R) over Minnesota Multiphasic Personality Inventory-2 Psychopathy Five (MMPI-2 PSY-5; with and without Aggressiveness Scale) for predicting aggression^a

Aggression Scales	MMPI-2 PSY-5 Scales ^b					MMPI2 PSY-5 Scales ^b + NEO-PI-R ^c										
	Ag	Co	Ps	NE	PE	Ag	Co	Ps	NE	PE	N	E	O	A	C	
<i>With PSY-5 Agressiveness</i>																
Physical Aggression	<i>0.48</i>	<i>-0.25</i>	<i>-0.05</i>	<i>0.09</i>	<i>-0.20</i>	<i>0.48</i>	<i>-0.21</i>	<i>-0.04</i>	<i>-0.03</i>	<i>-0.10</i>	<i>0.14</i>	<i>-0.03</i>	<i>-0.07</i>	<i>-0.10</i>	<i>-0.07</i>	
Verbal Aggression	<i>0.49</i>	<i>-0.12</i>	<i>-0.04</i>	<i>0.10</i>	<i>0.03</i>	<i>0.39</i>	<i>-0.01</i>	<i>-0.05</i>	<i>0.02</i>	<i>0.17</i>	<i>0.11</i>	<i>-0.05</i>	<i>0.10</i>	<i>-0.28</i>	<i>-0.10</i>	
Anger	<i>0.42</i>	<i>-0.06</i>	<i>-0.06</i>	<i>0.40</i>	<i>-0.14</i>	<i>0.45</i>	<i>0.06</i>	<i>-0.06</i>	<i>0.18</i>	<i>0.00</i>	<i>0.26</i>	<i>-0.03</i>	<i>0.00</i>	<i>-0.16</i>	<i>-0.16</i>	
Hostility	<i>0.16</i>	<i>-0.05</i>	<i>0.04</i>	<i>0.47</i>	<i>-0.26</i>	<i>0.26</i>	<i>-0.02</i>	<i>0.05</i>	<i>0.13</i>	<i>-0.15</i>	<i>0.46</i>	<i>-0.05</i>	<i>-0.08</i>	<i>0.03</i>	<i>0.05</i>	
Total Score	<i>0.47</i>	<i>-0.16</i>	<i>-0.02</i>	<i>0.33</i>	<i>-0.20</i>	<i>0.50</i>	<i>-0.08</i>	<i>-0.01</i>	<i>0.09</i>	<i>-0.04</i>	<i>0.31</i>	<i>-0.06</i>	<i>-0.04</i>	<i>-0.13</i>	<i>-0.09</i>	
<i>Without PSY-5 Agressiveness</i>																
Physical Aggression	–	<i>-0.38</i>	<i>0.10</i>	<i>0.22</i>	<i>-0.11</i>	–	<i>-0.31</i>	<i>0.05</i>	<i>0.17</i>	<i>-0.04</i>	<i>0.02</i>	<i>0.02</i>	<i>-0.05</i>	<i>-0.32</i>	<i>0.01</i>	
Verbal Aggression	–	<i>-0.25</i>	<i>0.11</i>	<i>0.23</i>	<i>0.13</i>	–	<i>-0.10</i>	<i>0.03</i>	<i>0.17</i>	<i>0.22</i>	<i>0.01</i>	<i>-0.01</i>	<i>0.12</i>	<i>-0.47</i>	<i>-0.04</i>	
Anger	–	<i>-0.17</i>	<i>0.07</i>	<i>0.51</i>	<i>-0.06</i>	–	<i>-0.04</i>	<i>0.03</i>	<i>0.36</i>	<i>0.06</i>	<i>0.15</i>	<i>0.02</i>	<i>0.02</i>	<i>-0.37</i>	<i>-0.09</i>	
Hostility	–	<i>-0.10</i>	<i>0.08</i>	<i>0.52</i>	<i>-0.23</i>	–	<i>-0.08</i>	<i>0.09</i>	<i>0.24</i>	<i>-0.10</i>	<i>0.40</i>	<i>-0.04</i>	<i>-0.06</i>	<i>-0.11</i>	<i>0.07</i>	
Total Score	–	<i>-0.29</i>	<i>0.13</i>	<i>0.46</i>	<i>-0.10</i>	–	<i>-0.19</i>	<i>0.08</i>	<i>0.30</i>	<i>0.02</i>	<i>0.18</i>	<i>-0.01</i>	<i>-0.02</i>	<i>-0.36</i>	<i>0.00</i>	

^a Significant beta weights are italicized ($P < 0.05$).

^b MMPI2-PSY-5 Scales: Ag, Aggressiveness; Co, Constraint; Ps, Psychoticism; NE, Negative Emotionality; PE, Positive Emotionality.

^c NEO-PI-R Scales: N, Neuroticism; E, Extraversion; O, Openness to Experience; A, Agreeableness; C, Con-

for all five AQ scales and ranged from 0.55 to 0.76 (see top portion of Table 4). With the five NEO-PI-R scales added to the multiple regression equation in the second step of the analyses, adjusted multiple R s were significant for all five AQ scales and ranged from 0.61 to 0.81, and the incremental validity of the NEO-PI-R scales over the MMPI-2 PSY-5 scales was also significant for all AQ scales, with changes in the R^2 ranging from 0.02 to 0.09.

With the PSY-5 Aggressiveness scale dropped from the regression analyses (see lower portion of Table 4), the adjusted multiple R s in the first step ranged from 0.38 to 0.68. Thus, the other four PSY-5 scales were still able to account for a fairly large and statistically significant amount of the variance in each of the five AQ scales. With the five NEO-PI-R scales added in the second step, adjusted multiple R s ranged from 0.56 to 0.74 and were statistically significant for all five AQ scales. Not surprisingly, without the PSY-5 Aggressiveness scale in the analyses the five NEO-PI-R domain scales were able to add a substantial amount of variance explained in the second step, with changes in the R^2 ranging from 0.07 to 0.16 (all statistically significant).

Table 5 presents the beta weights associated with the hierarchical regression analyses involving the PSY-5 scales in the first step and the NEO-PI-R scales in the second. Broadly speaking, beta weights from these analyses indicated that the NEO-PI-R Neuroticism and Agreeableness domain scales and the PSY-5 Aggressiveness scales were most important in the prediction of the five AQ scales, such that higher aggression was associated with higher neuroticism and aggressiveness and lower agreeableness. PSY-5 Constraint and Positive Emotionality also proved to be somewhat important in explaining the variance in the aggression scales, especially in the analyses in which the PSY-5 Aggressiveness scale was removed (see lower half of Table 5).

Table 6

Incremental validity of Minnesota Multiphasic Personality Inventory-2 Psychopathy Five (MMPI-2 PSY-5; with and without Aggressiveness Scale) over Revised NEO Personality Inventory (NEO-PI-R) for Predicting Aggression^a

Aggression Scale	NEO-PI-R		NEO-PI-R + MMPI-2 PSY-5		Incremental validity	
	Multiple R	$F(5,229)$	Multiple R	$F(10,224)$	R^2 Change	$F(5,224)$ Change
<i>With PSY-5 Aggressiveness</i>						
Physical Aggression	0.50	16.32*	0.67	18.72*	0.19	15.80*
Verbal Aggression	0.52	17.85*	0.61	14.78*	0.10	8.69*
Anger	0.64	33.34*	0.75	29.99*	0.15	15.73*
Hostility	0.69	43.24*	0.74	29.36*	0.07	8.37*
Total Score	0.68	40.53*	0.81	44.84*	0.19	26.34*
			Multiple R	$F(9,225)$	R^2 change	$F(4,225)$ change
<i>Without PSY-5 Aggressiveness</i>						
Physical Aggression	—	—	0.58	14.07*	0.09	8.53*
Verbal Aggression	—	—	0.56	12.54*	0.04	4.52*
Anger	—	—	0.69	23.91*	0.07	7.39*
Hostility	—	—	0.73	29.70*	0.05	6.99*
Total Score	—	—	0.74	31.79*	0.08	11.45*

^a All Multiple R s are adjusted

While the results presented in Tables 4 and 5 are in line with Ben-Porath and Waller's (1992) recommendation concerning the demonstration of the incremental validity of the NEO-PI-R, it was deemed necessary to reverse the analyses to examine the incremental validity of the PSY-5 over and above the NEO-PI-R. Such an analysis provides a full understanding of the relationships among the personality and aggression scales. Thus, Tables 6 and 7 contain the results of another set of hierarchical regression analyses with the NEO-PI-R scales entered in the first step, the PSY-5 scales entered in the second, and the five AQ scales used as separate criteria. In the first step of the analyses with the five NEO-PI-R scales used as predictors (see top portion of Table 6), adjusted multiple R s were statistically significant for all five aggression scales and ranged from 0.50 to 0.69. This demonstrates that the NEO-PI-R scales as a set are able to explain a rather large portion of the variance in the five AQ scales. With the PSY-5 scales added in the second step, adjusted multiple R s are exactly the same as reported in the previous analyses in which the NEO-PI-R scales were added to the PSY-5 scales in the prediction of the aggression scales. This is because the same 10 scales are combined in the second step of both sets of analyses (compare Tables 4 and 6). However, the important new information in this set of analyses is provided in the columns labeled "incremental validity" in Table 6. The incremental validity of the PSY-5 scales over the NEO-PI-R scales was statistically significant and quite substantial for all AQ scales, with changes in R^2 ranging from 0.07 to 0.19.

With the PSY-5 Aggressiveness scale dropped from the analysis, the first step of the analysis (which includes the five NEO-PI-R scales) remains the same as that reported in the top portion of

Table 7

Beta weights for regression analyses examining the incremental validity of Minnesota Multiphasic Personality Inventory-2 Psychopathy Five (MMPI-2 PSY-5; with and without Aggressiveness Scale) over Revised NEO Personality Inventory (NEO-PI-R) for predicting aggression^a

Aggression Scales	NEO-PI-R Scales ^b					NEO-PI-R Scales ^b + MMPI2 PSY-5 Scales ^c									
	N	E	O	A	C	N	E	O	A	C	Ag	Co	Ps	NE	PE
<i>With PSY-5 Aggressiveness</i>															
Physical Aggression	<i>0.15</i>	0.02	0.03	<i>-0.45</i>	-0.05	0.14	-0.03	-0.07	-0.10	-0.07	<i>0.48</i>	<i>-0.21</i>	-0.04	-0.03	-0.10
Verbal Aggression	0.09	0.11	<i>0.13</i>	<i>-0.51</i>	-0.02	0.11	-0.05	0.10	-0.28	-0.10	<i>0.39</i>	-0.01	-0.05	0.02	<i>0.17</i>
Anger	<i>0.41</i>	0.05	0.01	<i>-0.42</i>	-0.03	<i>0.26</i>	-0.03	-0.00	<i>-0.16</i>	<i>-0.16</i>	<i>0.45</i>	0.06	-0.06	<i>0.18</i>	0.00
Hostility	<i>0.63</i>	-0.10	-0.04	<i>-0.19</i>	0.10	<i>0.46</i>	-0.07	-0.07	0.01	0.03	<i>0.24</i>	-0.02	0.05	0.14	<i>-0.13</i>
Total Score	<i>0.41</i>	0.01	0.03	<i>-0.47</i>	0.01	<i>0.31</i>	-0.06	-0.04	<i>-0.13</i>	-0.09	<i>0.50</i>	-0.08	-0.01	0.09	-0.04
<i>Without PSY-5 Aggressiveness</i>															
Physical Aggression	—	—	—	—	—	0.02	0.02	-0.05	<i>-0.32</i>	0.01	—	<i>-0.31</i>	0.05	<i>0.17</i>	-0.04
Verbal Aggression	—	—	—	—	—	0.01	-0.01	0.12	<i>-0.47</i>	-0.04	—	-0.10	0.03	0.17	0.22
Anger	—	—	—	—	—	0.15	0.02	0.02	<i>-0.37</i>	-0.09	—	-0.04	0.03	<i>0.36</i>	0.06
Hostility	—	—	—	—	—	<i>0.40</i>	-0.04	-0.06	<i>-0.11</i>	0.07	—	-0.08	0.09	<i>0.24</i>	-0.10
Total Score	—	—	—	—	—	0.18	-0.01	-0.02	<i>-0.36</i>	0.00	—	<i>-0.19</i>	0.08	<i>0.30</i>	0.02

^a Significant beta weights are italicized ($P < 0.05$)

^b NEO-PI-R Scales: N, Neuroticism; E, Extraversion; O, Openness to Experience; A, Agreeableness; C, Conscientiousness

^c MMPI2-PSY-5 Scales: Ag, Aggressiveness; Co, Constraint; Ps, Psychoticism; NE, Negative Emotionality; PE, Positive Emotionality

Table 6. The important information here is listed in the bottom half of Table 6 in the “incremental validity” columns. Even with the Aggressiveness scale dropped from the equation, the remaining PSY-5 scales were able to add statistically significant levels of incremental validity, with changes in R^2 ranging from 0.04 to 0.09.

Table 7 contains beta weights associated with the hierarchical regression analyses just described. Of course, the only real difference between the beta weights reported here and those in Table 5 are those in the first step of the analyses with the NEO-PI-R alone in the prediction of aggression (see beta weights listed at top left of Table 7). Not surprisingly, these beta weights indicate that Agreeableness and Neuroticism were most important in the prediction of the aggression scales.

4. Discussion

The results obtained in the present study clearly indicate that the NEO-PI-R domain scales were able to add to the prediction of clinically relevant criteria (i.e. facets of aggression) over and above a measure commonly used in clinical assessment batteries (i.e. the MMPI-2-based PSY-5 scales) as called for in the literature by Ben-Porath and Waller (1992). This was evidenced by the hierarchical regression analyses with the PSY-5 scales in the first step and the NEO-PI-R scales in the second step, and was apparent even with the PSY-5 Aggressiveness scale included in the analyses. When the hierarchical analyses were reversed and the incremental validity of the PSY-5 scales over the NEO-PI-R scales was tested, it is not surprising that the more psychopathologically-oriented PSY-5 scales were able to add a substantial amount of variance to the prediction of different facets of aggression. Correlations and beta weights indicated that the NEO-PI-R scales of Neuroticism and Agreeableness, and the PSY-5 scales of Aggressiveness and Negative Emotionality were most important in the prediction of aggression. It is interesting to note that PSY-5 Negative Emotionality and NEO-PI-R Neuroticism were more highly related to the more dispositional aggression facets of anger and hostility as was expected. PSY-5 Constraint, Positive Emotionality, and Psychoticism were also somewhat important in terms of relationships with aggression.

Although empirically speaking incremental validity was demonstrated for the NEO-PI-R in the prediction of aggression, the importance of the incremental validity is nonetheless debatable. More specifically, the present findings still do not resolve the conflict over whether or not normal personality instruments should be used in clinical settings. Even if a clinician was handed a myriad of research findings demonstrating the incremental validity of normal-range measures such as the NEO-PI in the prediction of clinically-relevant criteria, he or she would still have to decide if the additional information obtained is enough to warrant the extra time needed to administer such an instrument (Ben-Porath & Waller, 1992). While the present study adds a small contribution to data on the incremental validity of normal personality instruments, clearly much more research needs to be conducted with other criteria to add to this literature. The decision to use such instruments is ultimately left to individual clinicians and if they have more data at their disposal their decisions will be better informed. Certainly the present study lends validity to Harkness and McNulty's (1994) Personality Psychopathology Five dimensional system and the use of the MMPI-2 PSY-5 in the assessment of basic psychopathologically oriented personality traits.

Beyond the contribution of the present study to the literature on the use of normal personality instruments in clinical settings, the results provide important information on the relationship between basic personality traits and aggression. As mentioned previously, theorists view aggression in different ways. Some psychologists tend to view aggression as a behavioral phenomenon contingent on situational factors, while others view it from a dispositional perspective (Geen & Donnerstein, 1998). Within the dispositional perspective, some have conceptualized aggression as a disposition in and of itself (e.g. Buss & Perry, 1992), while others claim aggressive behavior is determined by a combination of other dispositions (e.g. Caprara, 1986). Harkness et al. (1995) theorize that Aggressiveness is a basic personality trait within their Personality Psychopathology Five dimensional model. The present results show that several basic traits (especially neuroticism/negative emotionality and disagreeableness) are related to various aspects of aggression. Whether aggressive behavior is determined by a single disposition or is multiply determined by several dispositions is still a matter of theoretical debate.

The present results have other practical implications for clinicians that are more specifically related to aggression. Much research has been conducted which has attempted to identify important personality traits that differentiate violent/aggressive individuals from nonviolent, especially in prison settings (e.g. Selby, 1984) and with juvenile offenders (e.g. Berman & Paisey, 1984). Data from the present study suggest that several personality traits are important for predicting aggression, especially negative emotionality/neuroticism and disagreeableness. Armed with this knowledge clinicians and counselors may be better able to treat or prevent violent/aggressive behavior in a number of settings. For example, such knowledge may aid therapists in prison settings in terms of treatment and/or rehabilitation of violent criminals. Clinicians and counselors may also find this information to be helpful in the treatment and prevention of spousal and child abuse. Finally, school counselors and other therapists armed with such information will know what personality characteristics to be mindful of and may be able to act early to prevent patterns of violence throughout adolescence that can escalate in early adulthood. This final point seems particularly relevant given the recent acts of violence in schools on the part of adolescents across the United States.

The study reported here is not without limitations. One obvious limitation concerning the sample is the narrow age range of the participants. It is not certain that the same relationships between personality traits and aggression would hold up if a sample with a wider age range was used. Another potential limitation of the study is the fact that a clinical sample was not used in addition to the “normal” college-aged sample. Although it was appropriate to use a normal sample in the present study because a dimensional approach to personality psychopathology was being examined, it would be interesting to examine the same relationships using a sample with various diagnosed psychiatric disorders. Additionally, given the focus on aggression it would also have been interesting to conduct a similar investigation using samples of violent criminals or juvenile delinquents. Perhaps future research could address this issue. Finally, although a number of interesting relationships were observed between aggression and basic personality traits in this study, the measure of aggression used was more or less a dispositional measure of aggression as opposed to a behavioral measure. While it could be argued that the physical and verbal aggression facets measured by the AQ are more behaviorally oriented, future research could focus on determining how basic personality traits are related to aggressive behavior as assessed through behavioral frequency, observed behavior, and outcome data.

In conclusion, we feel that the present study makes a contribution to the literature on the use of normal-range personality instruments in clinical contexts, although practically speaking the use and added benefit of such instruments still seems open to debate. Hopefully, future research will shed more light on this issue and provide clinicians with enough evidence to make informed decisions on the matter.

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